

Workshop on Computational Models of Human Road User Behavior for Evaluation of Automated Vehicles

September 15, 2026, Naples, Stazione Marittima, Room Calipso

Abstracts of contributed talks and most recent schedule are available at <https://roadusermodels.github.io/>

Session	Start	End	Duration (min)	Item Type	Speaker(s)	Affiliation	Presentation Title / Topic
Session 1. Chair: Arkady Zgonnikov	8:30	8:40	10	Logistics	Arkady Zgonnikov	TU Delft, Organizer	Welcome, Workshop Objectives & Agenda
	8:40	9:10	30	Keynote	Biagio Ciuffo	European Commission – JRC	Driving behaviour models to operationalize the international regulatory framework for AV safety approval
	9:10	9:40	30	Keynote	Hiroki Nakamura & Sou Kitajima	Japan Automobile Research Institute (JARI)	Defining Safety Boundaries for Automated Driving Systems: From real traffic data to computational human road-user models
Session 2. Chair: Jonas Bärgman	9:40	10:00	20	Contributed	Kevin Kefauver	VTTI	Prudent Attentive Sober (PAS) Driver Definition in US National Crash Data and its Application to Human Behavioral Models
	10:00	10:20	20	Contributed	Arturo Tejada Ruiz	TNO	Bending the (traffic) rules, competent driver style!
	10:20	10:40	20	Contributed	Giovanni Albano	Univ. of Naples Federico II	Risk perception and braking reaction model of competent and careful human drivers in safety-critical cut-in scenarios
	10:40	11:00	20	Contributed	Federico Scari	RDW	A Scenario-Based Grading Framework for ADS/ADAS Consumer-Testing Campaigns Benchmarked Against a Probabilistic Human Driver Model
Break	11:00	11:30	30	Coffee Break			
Session 3. Chair: Gustav Markkula	11:30	12:00	30	Keynote	Jonas Bärgman	Chalmers Univ. of Technology	Representativeness in ADS safety impact assessment (and training): baseline and traffic-agent challenges
	12:00	12:30	30	Keynote	Manel Hammouda	BMW Group	Cognition and Reward: A Dual Perspective on Driver Behavior Modeling
	12:30	12:50	20	Contributed	Yueyang Wang	Univ. of Leeds	Task-oriented scoring of driver behaviour models for automated driving systems evaluation
	12:50	12:53	3	Teaser talk	Malin Svärd	Volvo Cars	Quantification of Human Acceleration Behavior across Speeds Using Real-World Driving Data
	12:53	12:56	3	Teaser talk	Konstantinos Mattas	European Commission – JRC	Combining Evasive and Braking Reactions for Safety Reference Models in Automated Vehicles
	12:56	12:59	3	Teaser talk	Samir Mohammad	TU Delft	General-purpose LLMs as Models of Human Driver Behavior: The Case of Simplified Merging
Break	13:00	14:00	60	Lunch Break			
Session 4. Chair: Luigi Pariota	14:00	14:30	30	Keynote	Gustav Markkula	Univ. of Leeds	Mechanistic, machine-learned, and hybrid modelling of road user behaviour
	14:30	14:50	30	Keynote	Johan Engström	Waymo	SAE J3330: Toward a shared language for driving behavior reference models
Break	15:00	15:15	15	Break			
Session 5. Chair: Johan Engström	15:15	15:35	20	Contributed	Julian Frederik Schumann	TU Delft	Active Inference as a model of human driving behavior
	15:35	15:55	20	Contributed	Christian Rössert	cogniBIT GmbH	Beyond benchmark metrics: a cognitive driver model as a behavioral reference for AV evaluation
	15:55	16:15	20	Contributed	Saeed Rahmani	TU Delft	Drive with Style: Style-Conditioned Self-Play for Behaviorally Diverse Traffic Simulation
	16:15	16:35	20	Contributed	Arash Tavakoli	Villanova Univ.	Driver World Models: Unified Representations of Drivers in Semi-Automated Driving
Break	16:35	16:45	10	Break			
Session 6. Chair: Shu-Yuan Liu	16:45	17:35	50	Discussion	Shu-Yuan Liu, Arkady Zgonnikov	Moderated by Organizers	Interactive Breakout Groups & Plenary Synthesis
	17:35	17:45	10	Logistics	Arkady Zgonnikov	Organizing Committee	Closing Remarks